

An Introduction to 3D Printing

Part 3: Science

*Vibrations and Waves, Linear Motion, and Thermal
Expansion*

Outline

Part 1: Printers

Process

Limitations

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

Thermal Expansion

Vibrations and Waves

- Tensioned belts (and pulleys and bearings) connect motors and the toolhead.
- The printer creates vibrations when the toolhead:
 - Speeds up.
 - Slows down.
 - Changes direction.
- Belts (strings) + vibrations (waves) = Physics.
 - Motions systems with belts have a natural frequency.
 - What does this mean?
 - *Hint:* Think of a guitar or piano string.

Repeating Mistakes

- What happens when the toolhead hits a natural frequency?
 - Resonance.
- Toolhead excites the belts (adds energy) with vibrations.
- Resonance maximizes vibrations.
 - Standing waves undergo constructive interference.

Repeating Mistakes

- Vibrations make the belts push and pull on the toolhead.
- This creates small errors in nozzle position.
- **Errors repeat based on the frequency.**
- Imagine trying to write on paper with someone bumping your arm at a constant time interval.

- Where are the repeating errors?
- Sharp corners.
- Which direction was the toolhead moving?
- From left to right.

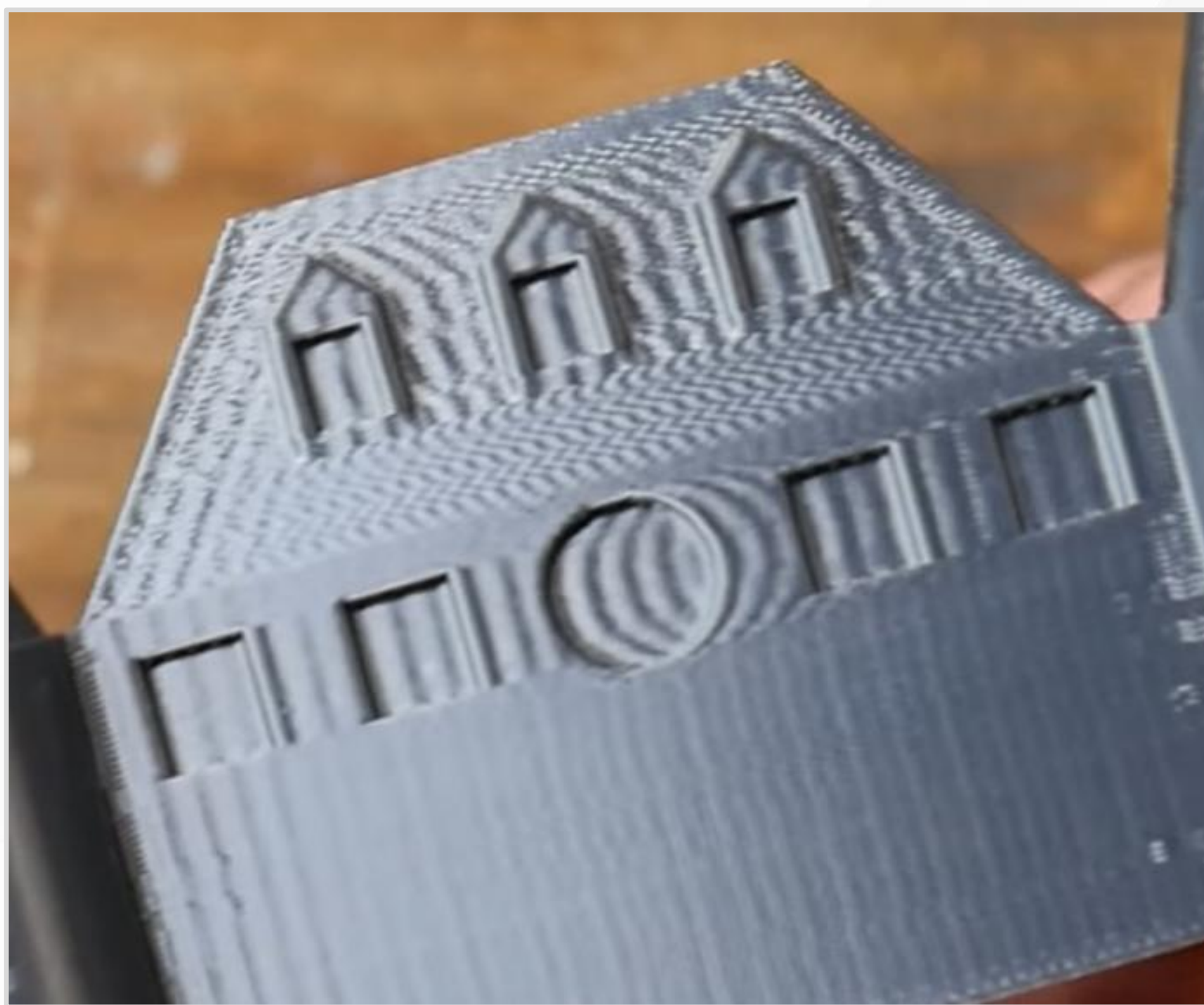
(Screenshot from
Teaching Tech.)



- How can you determine frequency?
- Lookup printing speed in settings.
- Measure distance between repeating patterns.

$$freq = \frac{print_speed}{repeating_dist}$$

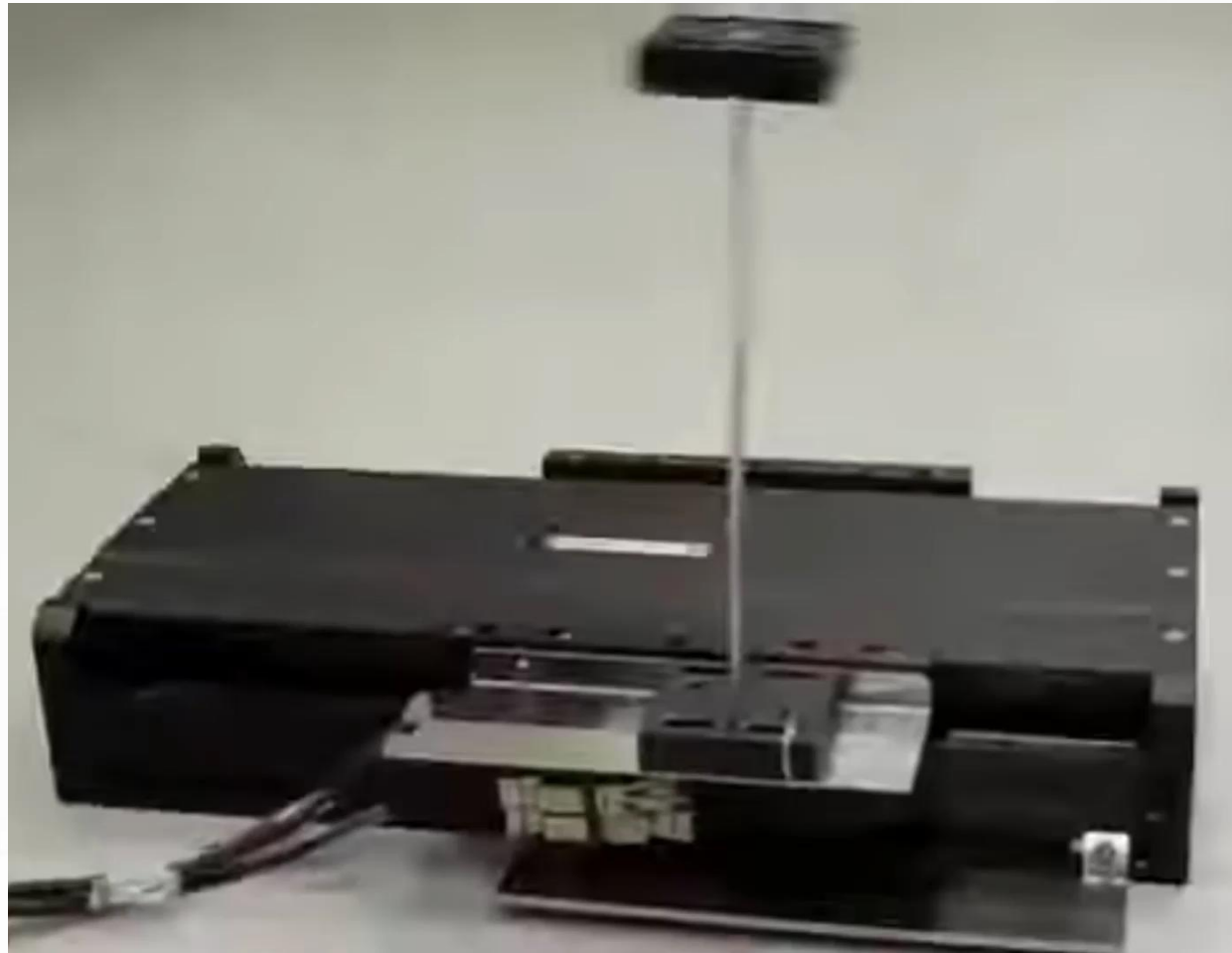
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Teaching Tech.)



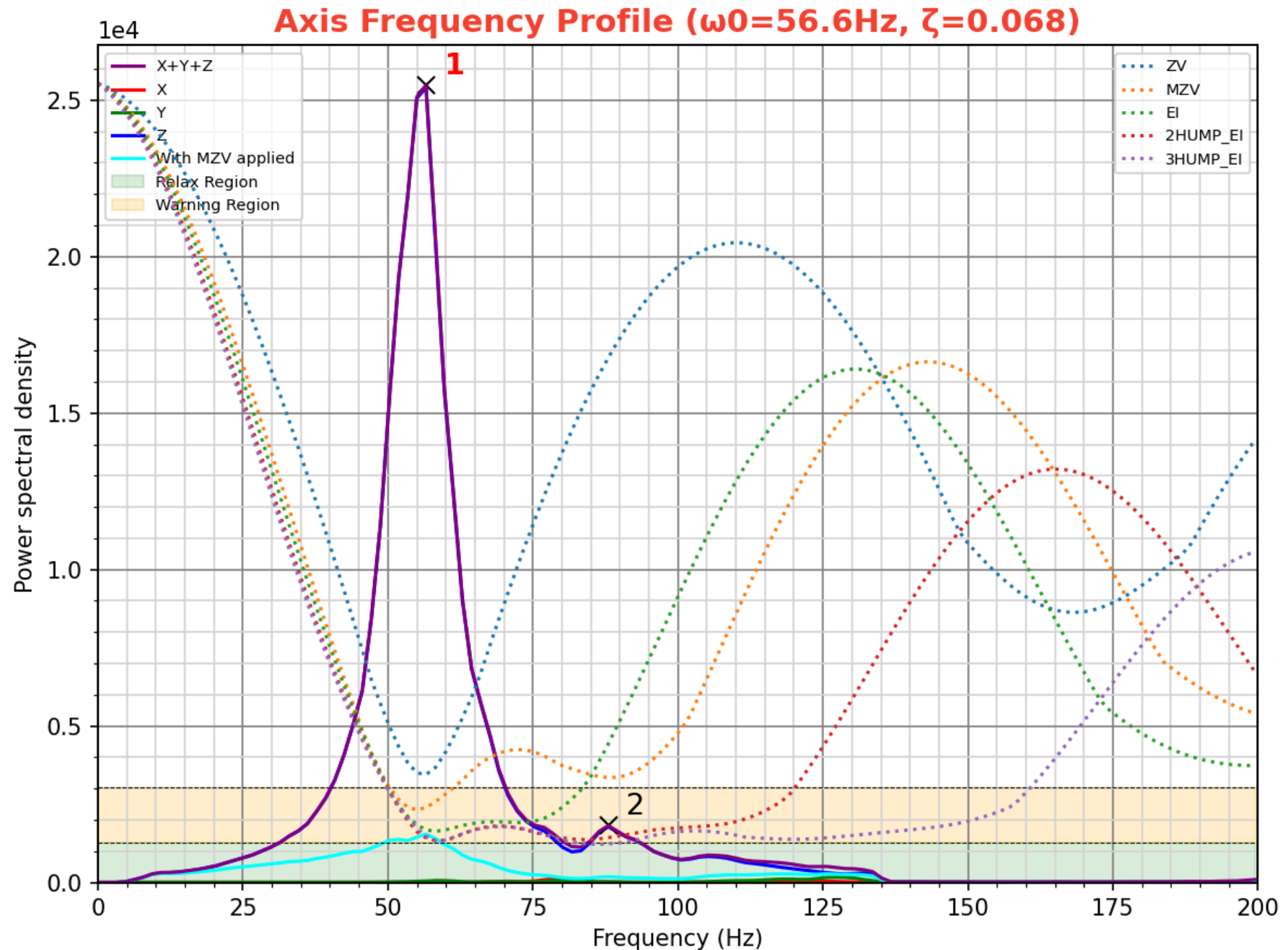
Input Shaping

- How can you fix vibrations?
- Input shaping adds extra movements that cancel vibrations.
 - Destructive wave interference.
- Similar to noise-canceling headphones.

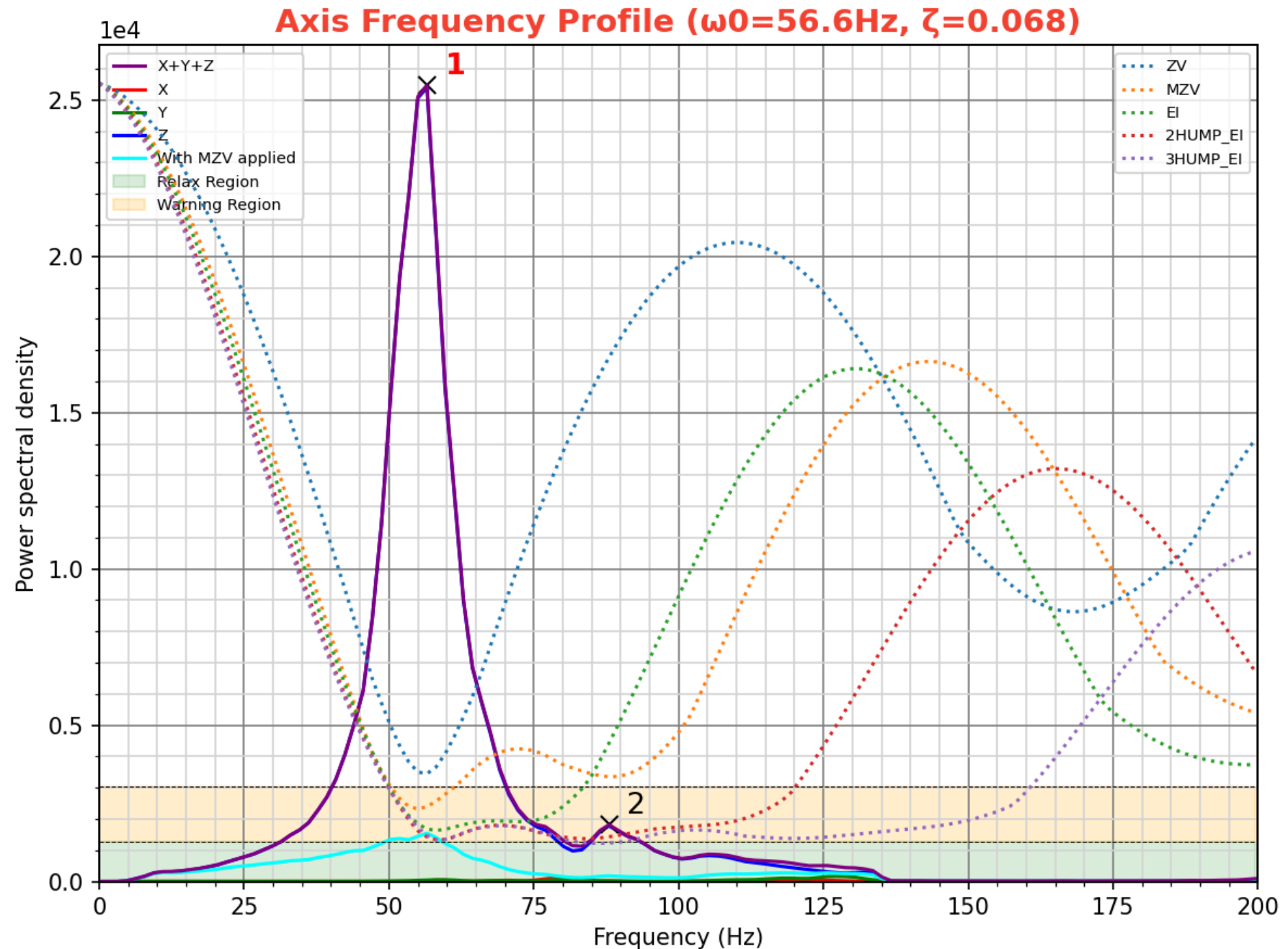
(Demo by ACS Motion Control.)



- Shake and Tune Plugin.
- Attach an accelerometer to the toolhead.
- Move the toolhead back and forth to sweep frequencies.
- Script analyzes the results, makes this graph, and suggests Input Shaper settings.



- X axis results for my printer.
- Single and narrow peak is ideal.
- **Purple** → Before
- **Cyan** → After
- Ignore high frequencies on the right side of the graph.
- Each printer and each axis has a characteristic shape.



Speed Limits

I want to print faster. Here is what I have:

- *Acceleration*: 10,000 mm/s²
- *Max speed*: 300 mm/s
- *Nozzle diameter*: 0.4 mm
- *Layer height*: 0.2 mm
- *Max volumetric flow*: 22 mm³/s
- Should I upgrade motors or hot end?
- What about top speed or acceleration?

Velocity and Acceleration

- What is the distance to reach top speed?
 - Starting speed (cornering speed, jerk): 5 mm/s
 - Increasing cornering speed *greatly* increases vibrations.
- What is the smallest extrusion that reaches top speed?
 - 9 mm to accelerate *and* decelerate.

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$300^2 = 5^2 + 2 \times 10,000(\textit{dist})$$

$$\textit{dist} = 4.5 \textit{ mm}$$

Velocity and Flow

Can my hotend melt enough plastic at top speed?

- Almost.

$$\textit{print_speed} = \frac{\textit{flow_rate}}{\textit{cross_section}}$$

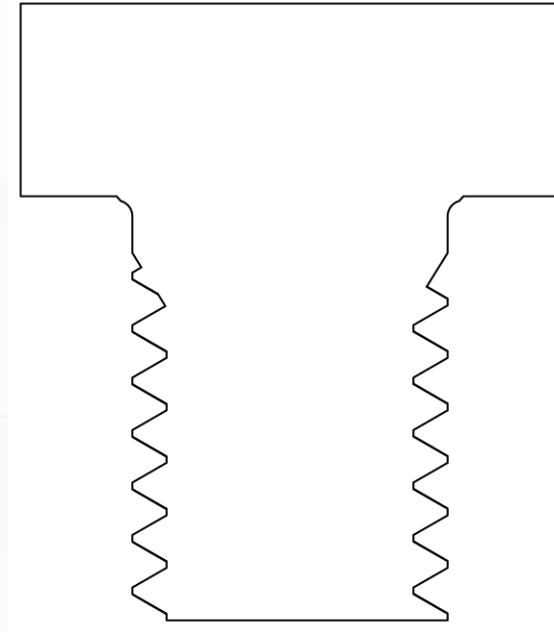
$$\textit{print_speed} = \frac{\textit{flow_rate}}{\textit{extrusion_width} \times \textit{layer_height}}$$

$$\textit{print_speed} = \frac{22}{0.4 \times 0.2}$$

$$\textit{print_speed} = 275 \text{ mm/s}$$

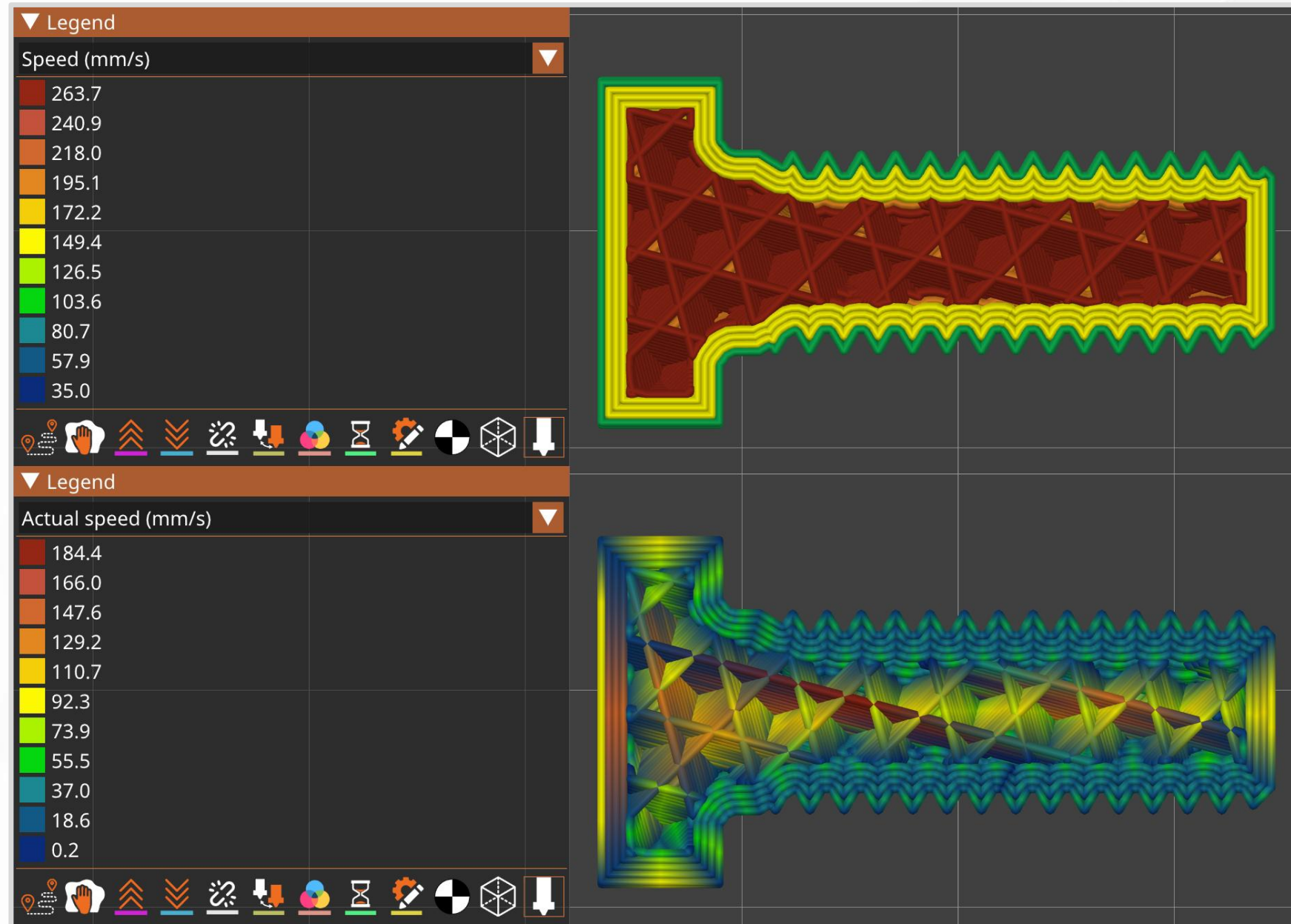
Finding Limits

- Real life limits are more complex and nuanced.
- *Example:* Printed bolt
 - 32 mm long (about 1 inch)
 - 8 mm diameter (about 3/8 inch).
- What is going to limit my printer?
- Will the printer hit top speed (about 275 mm/s)?
 - The dimensions are under 9mm ...
- Let PrusaSlicer do all the math!



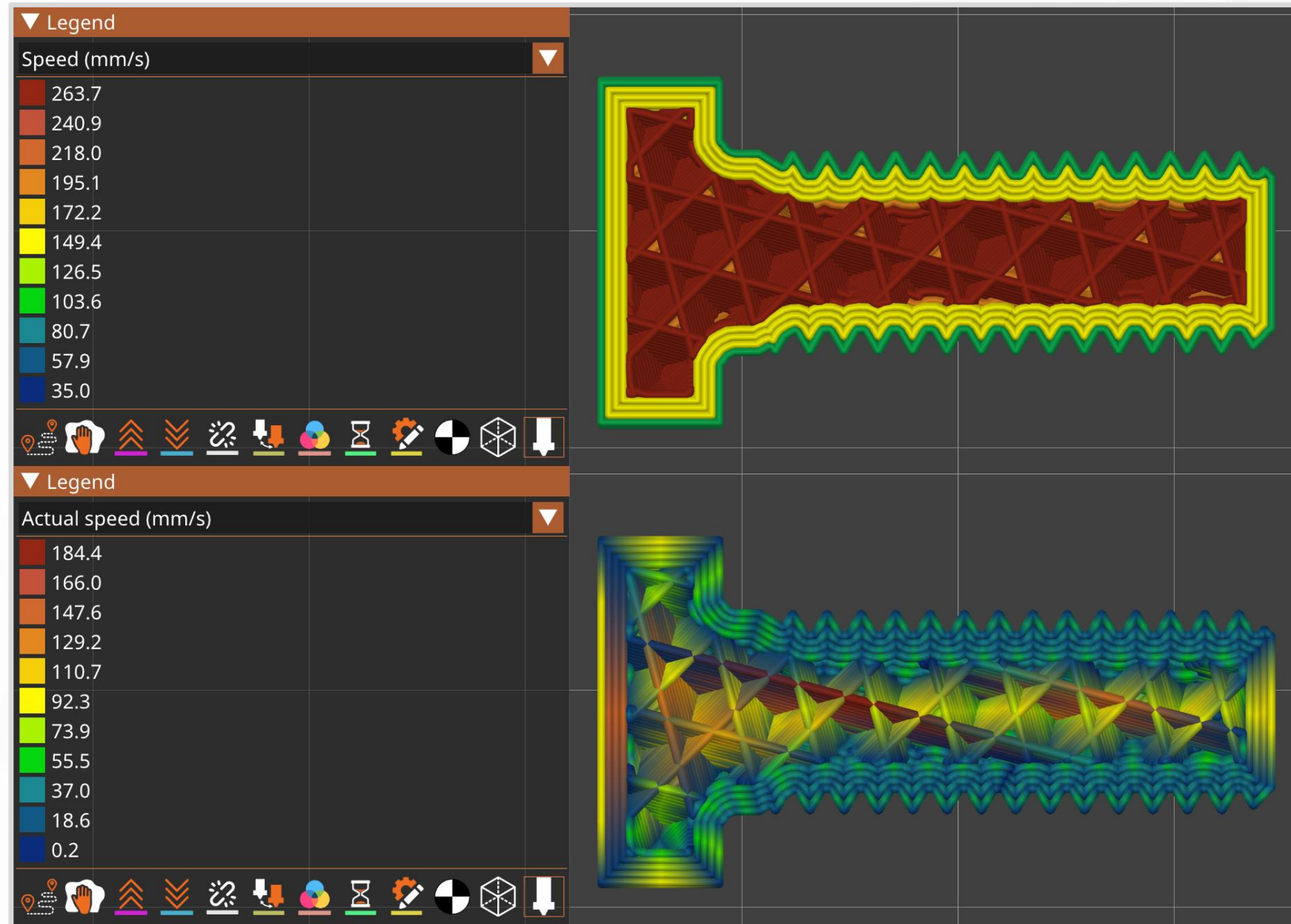
- Max requested speed: **264 mm/s**

- Max actual speed: **184 mm/s**



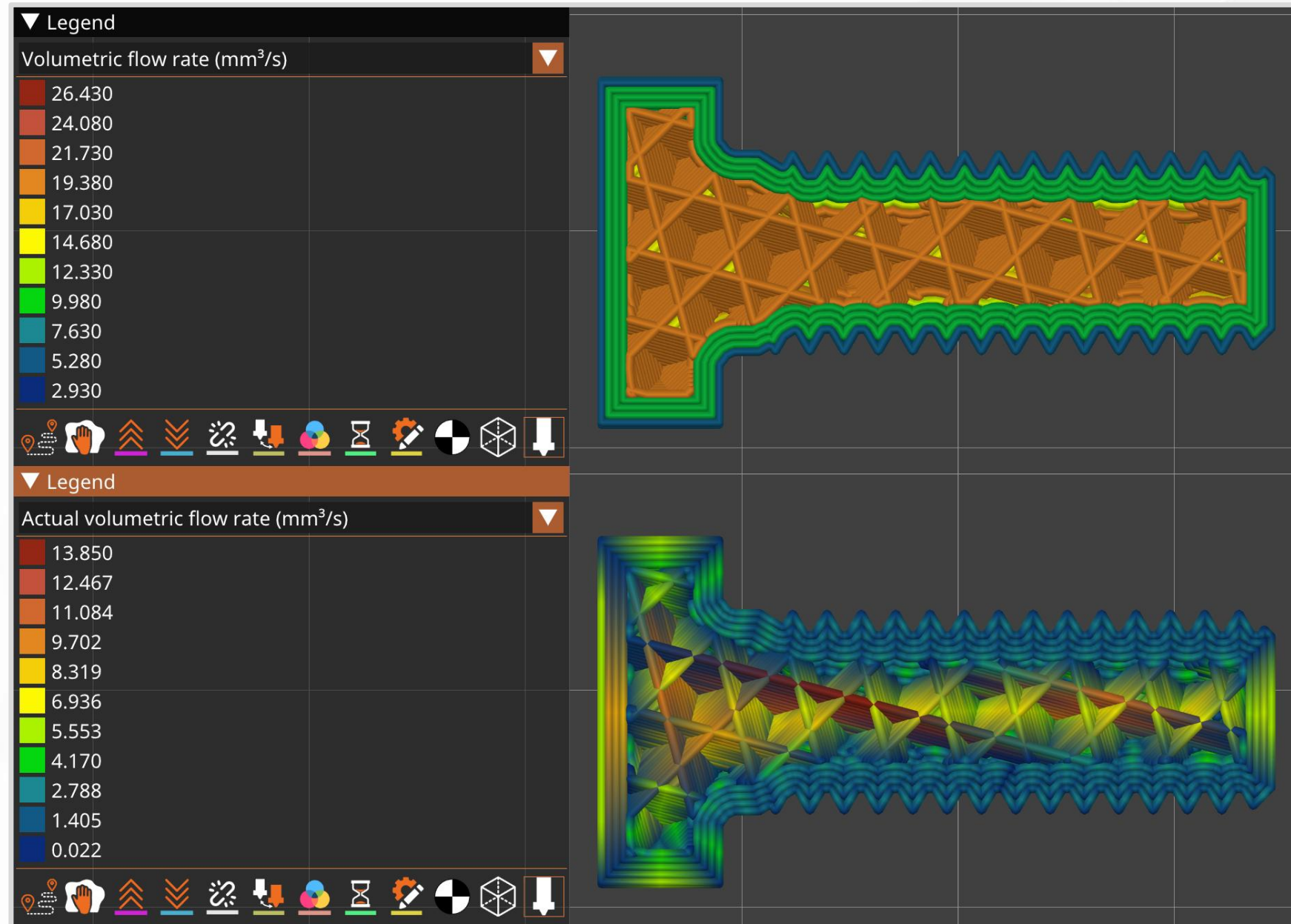
Why is the requested speed non-uniform?

- Adjust speeds according to geometry.
- Slow and pretty on the outside.
- Fast and rough on the inside.



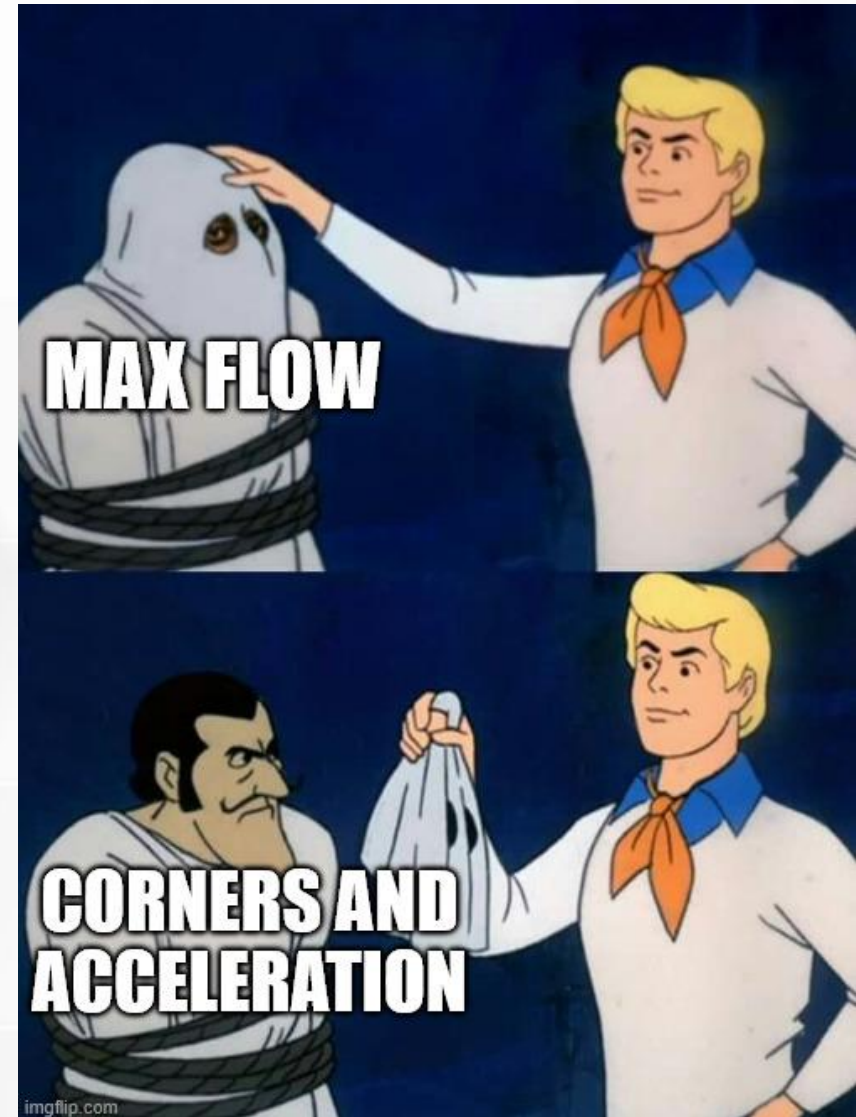
- Max requested flow:
22 mm³/s

- Max actual flow:
14 mm/s



Speed Limits

- Max requested speed: 264 mm/s
 - Close to estimate for flow-limited top speed.
- Max actual speed: 184 mm/s
- Max requested flow: 22 mm³/s
- Max actual flow: 14 mm³/s
- My printer does not hit max speed, or max flow.
- Limited by acceleration and geometry (corners).

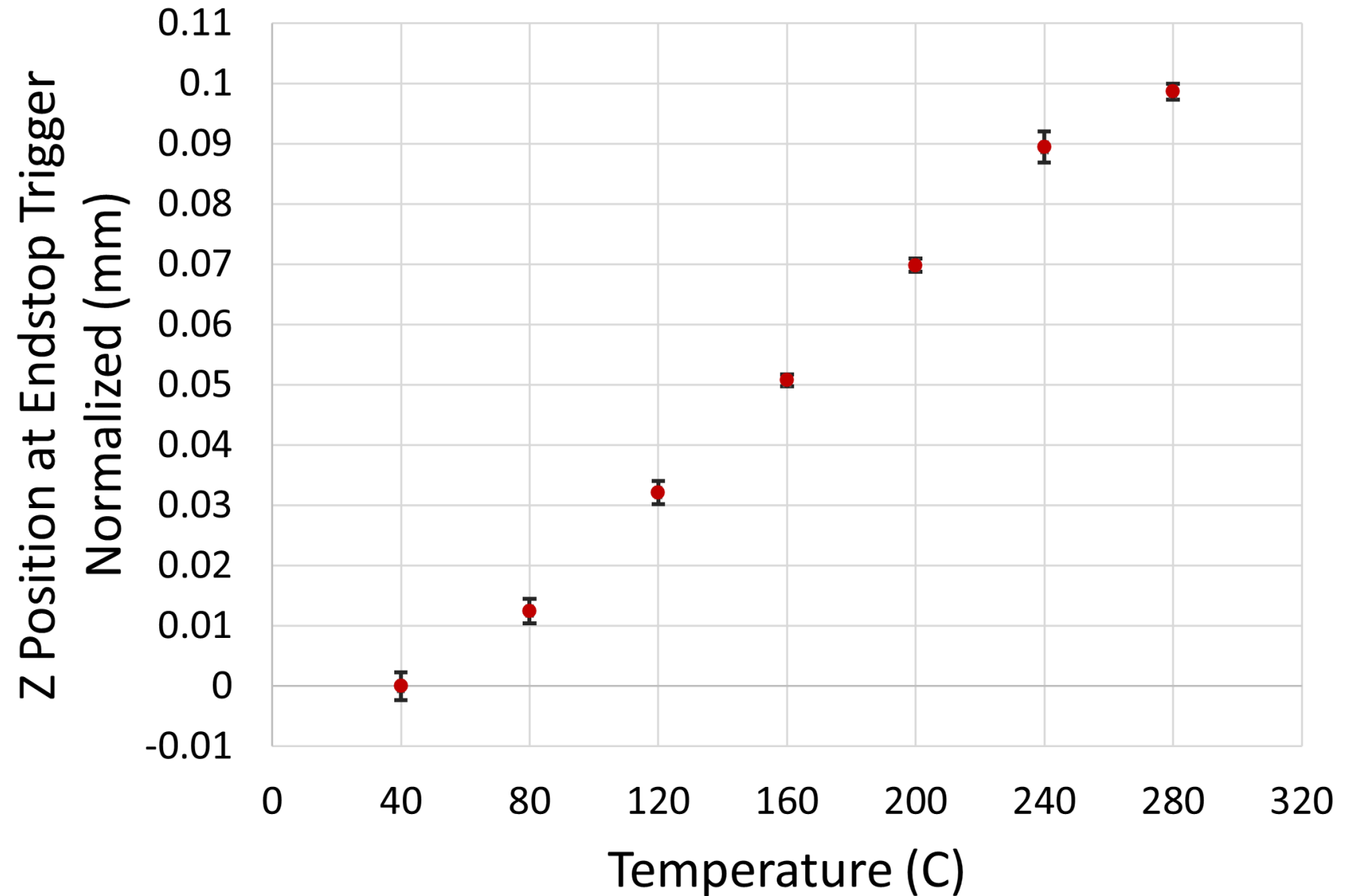


Thermal Expansion

- Metal expands as it heats.
 - Coefficient of linear expansion.
- Estimate brass nozzle using a 12mm cylinder:
 - $\Delta T = 150^{\circ}\text{C} \rightarrow +0.034 \text{ mm}$.
 - $\Delta T = 300^{\circ}\text{C} \rightarrow +0.068 \text{ mm}$.
 - Does not include effects from heatsink or heat break.
- Is this large enough to affect the first layer?
 - **Yes, first layers are very sensitive.**
 - Well known from experience.

- Data from my own printer.
- Set temperature and wait 10 minutes.
- 19 samples per point.
- My printer uses two probes and an offset calculation → normalized to zero.
- Error bars $\pm$ one standard deviation.

Thermal Expansion of a Brass Nozzle



Thermal Expansion

- What can you do about it?
 - “Heat soak.” Let everything expand and wait for equilibrium.
 - Then probe nozzle position.
 - First layer is over-squished if you skip waiting.
 - Print beds need this same treatment.

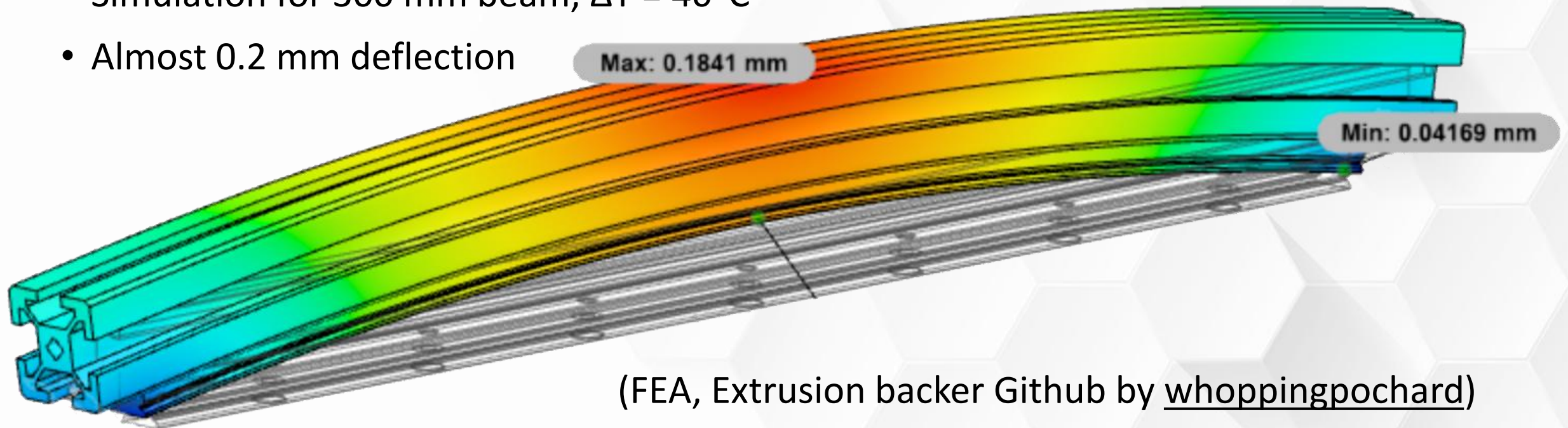
Thermal Expansion

- Printers have lots of parts.
- What happens when you assemble metals with different expansion rates?
 - Frame extusions
 - Sheet metal
 - Screws
 - Bearings
 - Rods and rails



(Bimetallic) Thermal Expansion

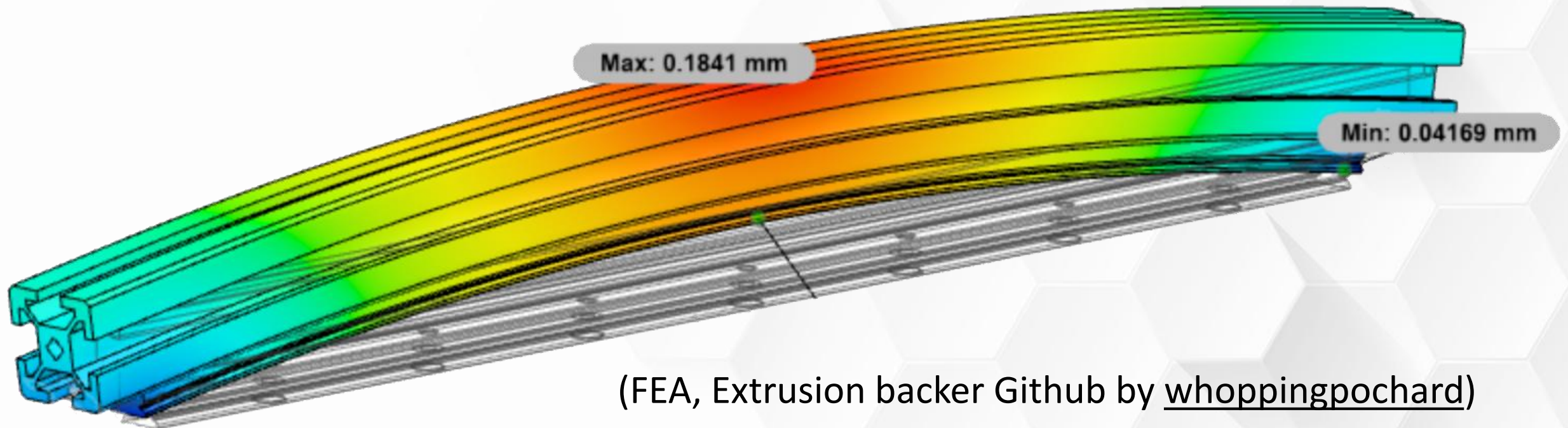
- Aluminum extrusion on top.
- Stainless-steel linear rail on bottom.
- Bimetallic “strips” **bend** and form an arc.
- Simulation for 300 mm beam, $\Delta T = 40^\circ\text{C}$
- Almost 0.2 mm deflection



(FEA, Extrusion backer Github by [whoppingpochard](#))

(Bimetallic) Thermal Expansion

- Old fashioned thermometers used bimetallic strips to make the dial.
- Which material has higher expansion coefficient?
- Aluminum because it is on the outside of the curve.



(FEA, Extrusion backer Github by [whoppingpochard](#))

